```
home-digital-twin-bucket/
├── sensors/
│   ├── config/sensor_id1/config.json
│   └── firmware/temp_humidity_esp32-v2.1.0.bin
├── data_exports/
│   ├── daily/2025-08-18/sensor_data.csv
│   └── monthly/2025-08/complete_export.json
├── 3d_models/
│   ├── mesh/home.obj
│   └── tiles/tileset.json
├── media/images/2025-08-18/water_meter_1.jpg
└── documents/
    ├── manuals/MHI_DXK18ZTLA-WF.pdf
    ├── maintenance/2024-01-02_aircon_service.pdf
    └── floorplans/home.png
```

4.8.3 Storing relationship data

For complex relationship data in my home digital twin, I need a database that handles interconnected information efficiently. Graph databases excel here over traditional relational databases by providing natural representation of connections, efficient traversal algorithms, flexible schema evolution, and rich contextual insights that reveal patterns difficult to infer from other data models.

Graph database options include fully managed cloud services (Amazon Neptune, Azure Cosmos DB, Neo4j AuraDB) and self-hosted solutions. While managed services suit large, mission-critical workloads, my small home digital twin model doesn't justify their cost. Instead, I host Memgraph in a Docker container on an AWS EC2 server, providing low-cost operation while maintaining elasticity and future portability to other compute services. In chapter 5, we will look at how I use this graph database to store relationship data in my home digital twin.

> ***NOTE***
>
> Docker is a platform that packages applications and their dependencies into lightweight, portable containers that run consistently across any environment. This eliminates compatibility issues and makes it easy to deploy complex applications.

4.9 Summary

- There are a number of different types of data that are found in most digital twins including timeseries, relational, object, spatial, and derived data which includes features and embeddings.
- Digital twins source data from both internal data stores, and from external sources.